

STA4322 Homework 2

Jordan Sutherland

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1. 7.11

$$P(|\bar{Y} - \mu| < 2) = P\left(\left|\frac{\bar{Y} - \mu}{4/\sqrt{9}}\right| < \frac{2}{4/\sqrt{9}}\right) = P(|z| < 1.5) = P(|z| \leq 1.5) \\ = \Phi(1.5) - \Phi(-1.5)$$

```
pnorm(1.5,0,1,lower.tail=FALSE)
```

```
## [1] 0.0668072
```

```
pnorm(1.5,0,1,lower.tail=TRUE)
```

```
## [1] 0.9331928
```

```
0.9331928-0.0668072
```

```
## [1] 0.8663856
```

The first line of R code solves for $\Phi(1.5)$. The middle line of R code solves for $\Phi(-1.5)$. The last line of R code calculates $\Phi(1.5) - \Phi(-1.5)$. So the probability of the sample mean being within 2 square inches of the population mean is 0.8664.

2. 7.12

$$\Phi(1.6448) - \Phi(-1.6448) = \Phi\left(\frac{1\sqrt{n}}{4}\right) - \Phi\left(-\frac{1\sqrt{n}}{4}\right) \\ \frac{\sqrt{n}}{4} = 1.6448 \quad \sqrt{n} = 6.5792 \quad n^2 = 43.2859$$

```
qnorm(0.95,0,1,lower.tail=TRUE)
```

```
## [1] 1.644854
```

The line of code calculates the Φ value such that $\Phi(a) - \Phi(-a) = 0.90$. Then the forester must measure 44 trees in order to ensure the sample mean is within 1 square inch of the population mean with probability 0.90.

3. 7.26

$$-t_{n-1,1-\alpha/2} \frac{s}{\sqrt{n}} \leq \bar{Y} - \mu \leq t_{n-1,1-\alpha/2} \frac{s}{\sqrt{n}} \quad -1.8595 \frac{2}{\sqrt{9}} \leq \bar{Y} - \mu \leq 1.8595 \frac{2}{\sqrt{9}} \\ -1.2397 \leq \bar{Y} - \mu \leq 1.2397$$

```
qt(0.95,8,lower.tail=TRUE)
```

```
## [1] 1.859548
```

The line of R code finds the critical constant $t_{n-1,1-\alpha/2}$. So for $P(g_1 \leq \bar{Y} - \mu \leq g_2)$, the two statistics are $g_1 = -1.2397$ and $g_2 = 1.2397$.

4. 7.42

$$\text{A) } P(\bar{Y} - 14 > 0.5) = P\left(\frac{\bar{Y}-14}{2/\sqrt{100}} > \frac{0.5}{2/\sqrt{100}}\right) = P(z > 2.5)$$

```
pnorm(2.5,0,1,lower.tail=FALSE)
```

```
## [1] 0.006209665
```

The line of R code calculates the area more than 2.5 standard deviations above the mean on a standard normal distribution.

$$\text{B) } \bar{Y} \pm z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} = \bar{Y} \pm 1.9600 \frac{2}{\sqrt{100}} = \bar{Y} \pm 0.392$$

```
qnorm(0.975,0,1,lower.tail=TRUE)
```

```
## [1] 1.959964
```

The line of R code calculates the Φ value such that $\Phi(a) - \Phi(-a) = 0.95$. An interval that includes the average fracture strength of 100 randomly selected pieces of glass with probability of 0.95 is $(\bar{Y} - 0.392, \bar{Y} + 0.392)$.